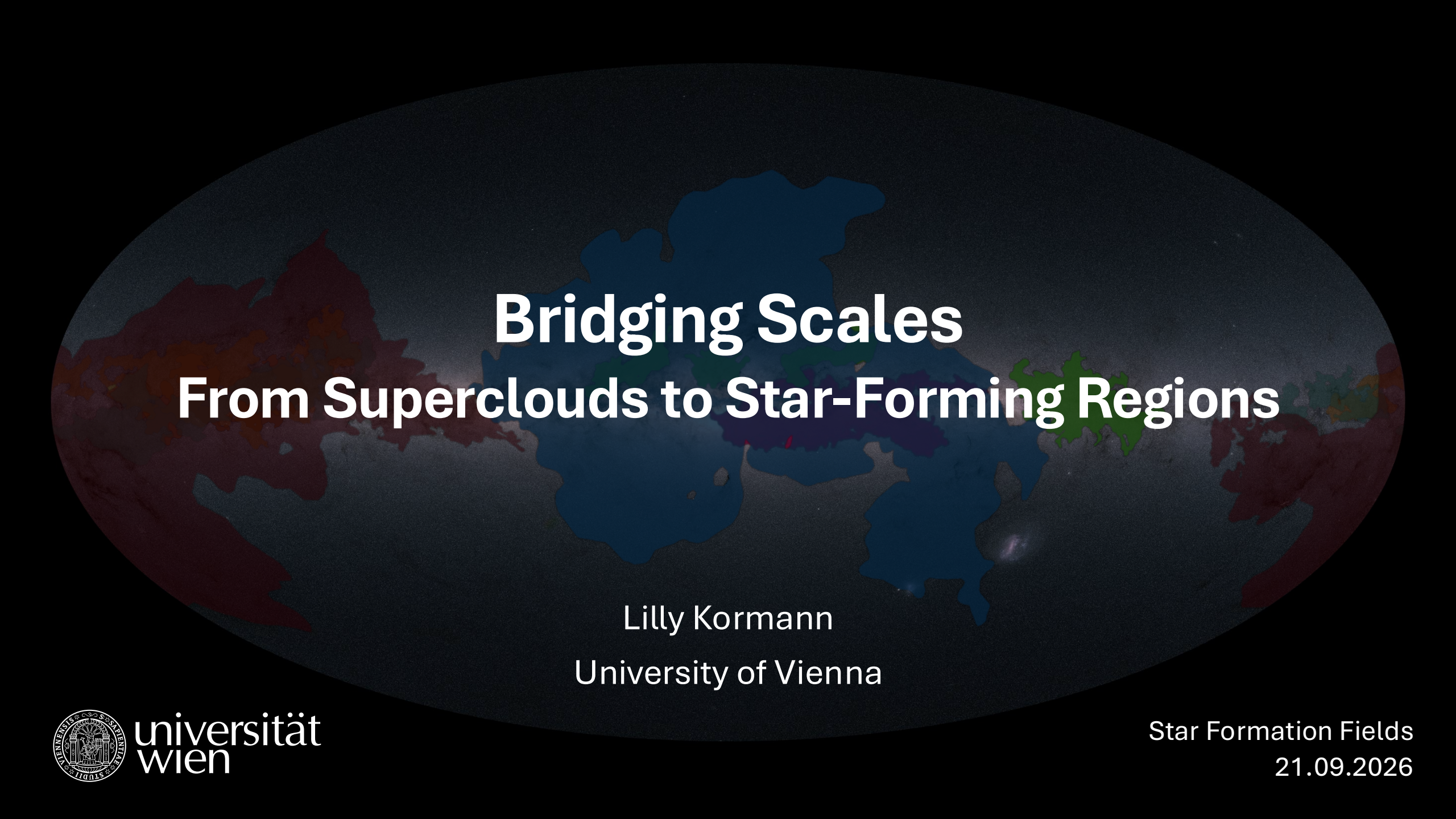




Bridging Scales

From SupercLOUDs to Star-Forming Regions

Lilly Kormann

University of Vienna

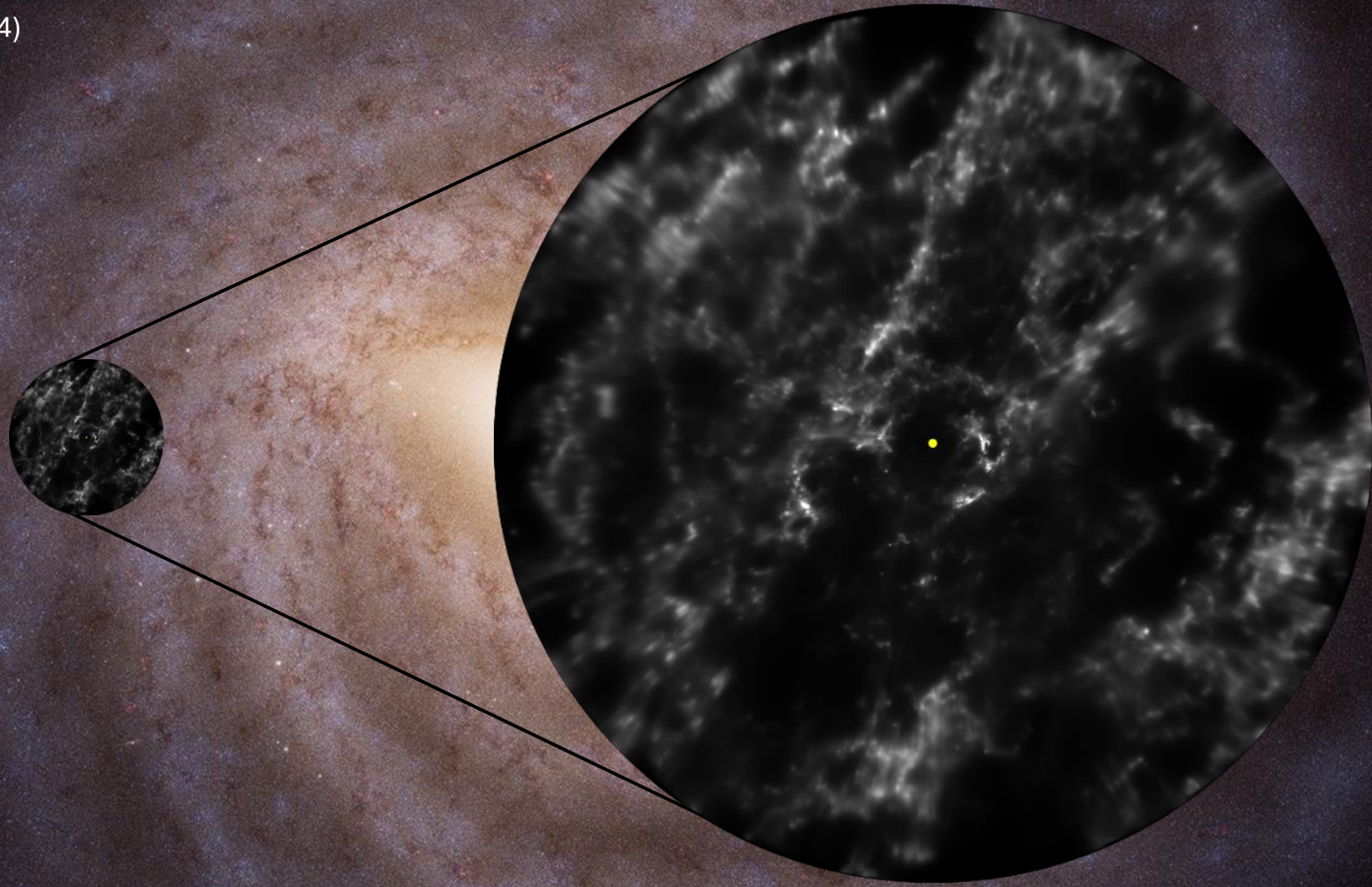
Early History of Superclouds

- some of the largest recognized structures within the interstellar medium
- first mentioned by Elmegreen & Elmegreen (1983)
 - neutral atomic hydrogen superclouds in spiral arms in external galaxies
 - masses between 10^6 and $10^7 M_{\odot}$ and kiloparsec lengths
- further work done by Elmegreen & Elmegreen (1987)
 - identified neutral atomic hydrogen superclouds in the inner Galaxy
 - noted spatial correlation with giant molecular cloud complexes

Artistic Impression of the Milky Way

ESA/Gaia/DPAC, Stefan Payne-Wardenaar

Dust map: Edenhofer et al. (2024)



Finding Superclouds

previously known:

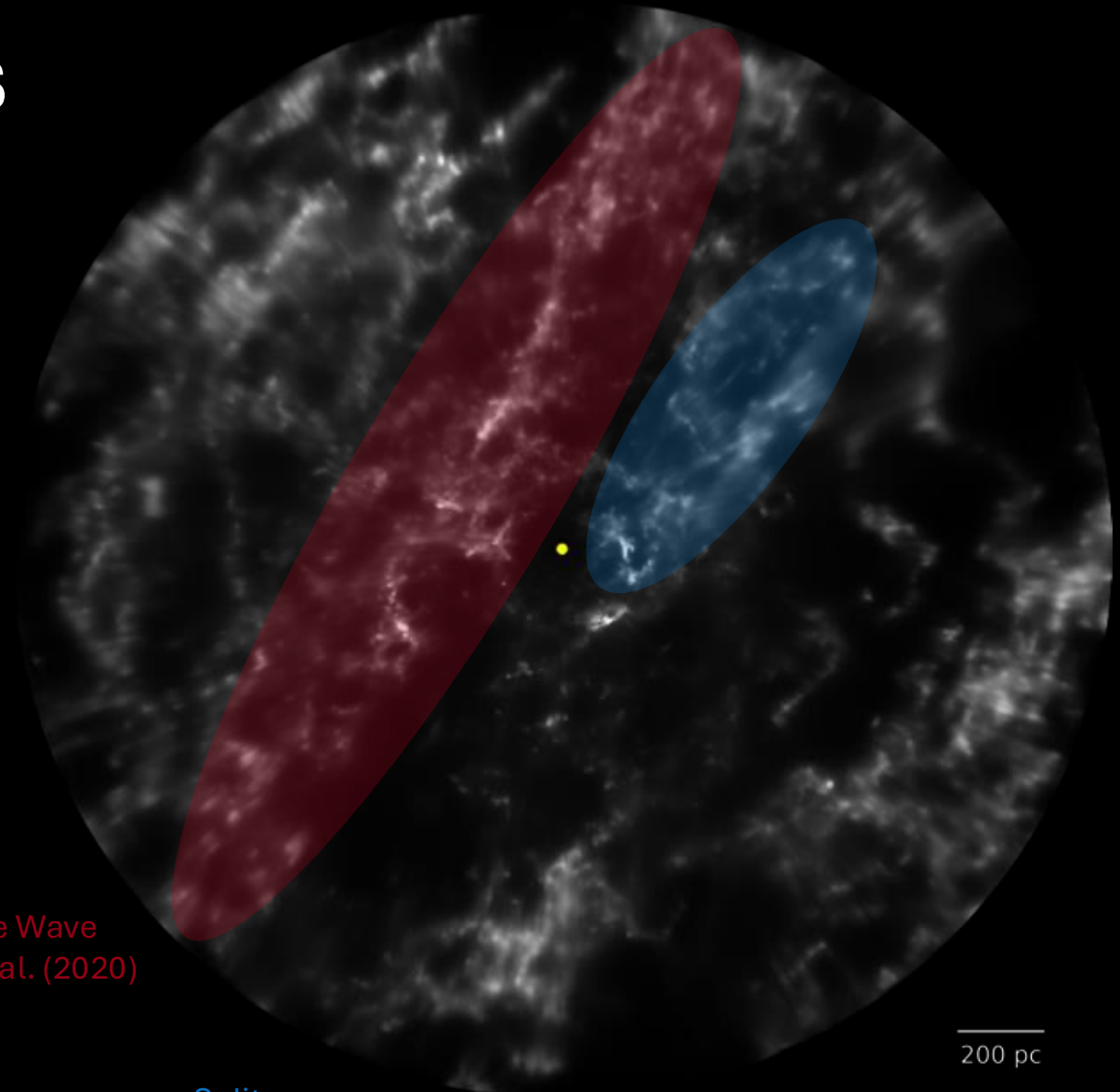
Radcliffe Wave

Split

HOP structure finder from
Colman et al. (2024)

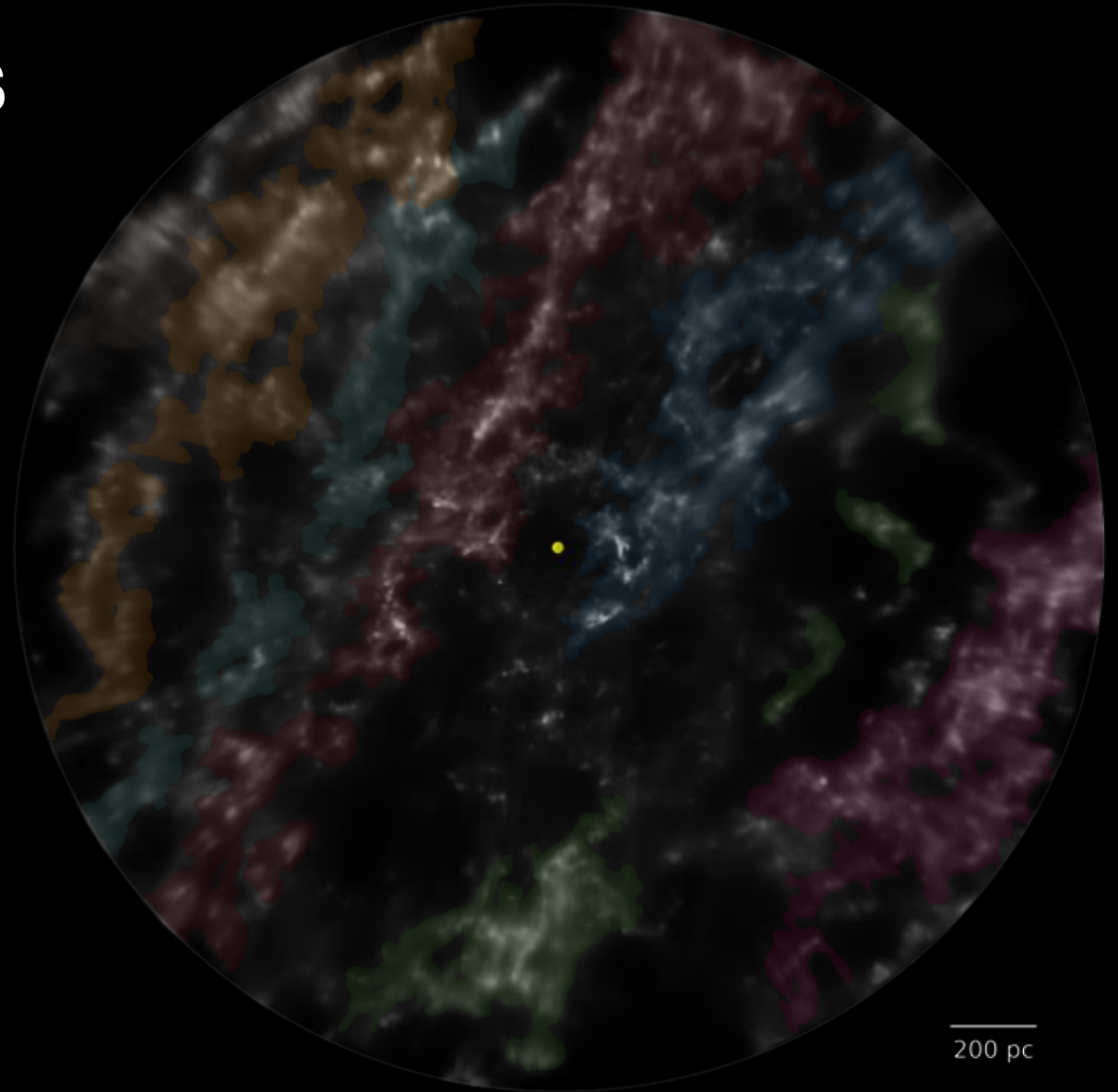
Radcliffe Wave
Alves et al. (2020)

Split
Lallement et al. (2019)



Finding Superclouds

HOP structure finder from
Colman et al. (2024)



Superclouds

7 superclouds of the local
Milky Way

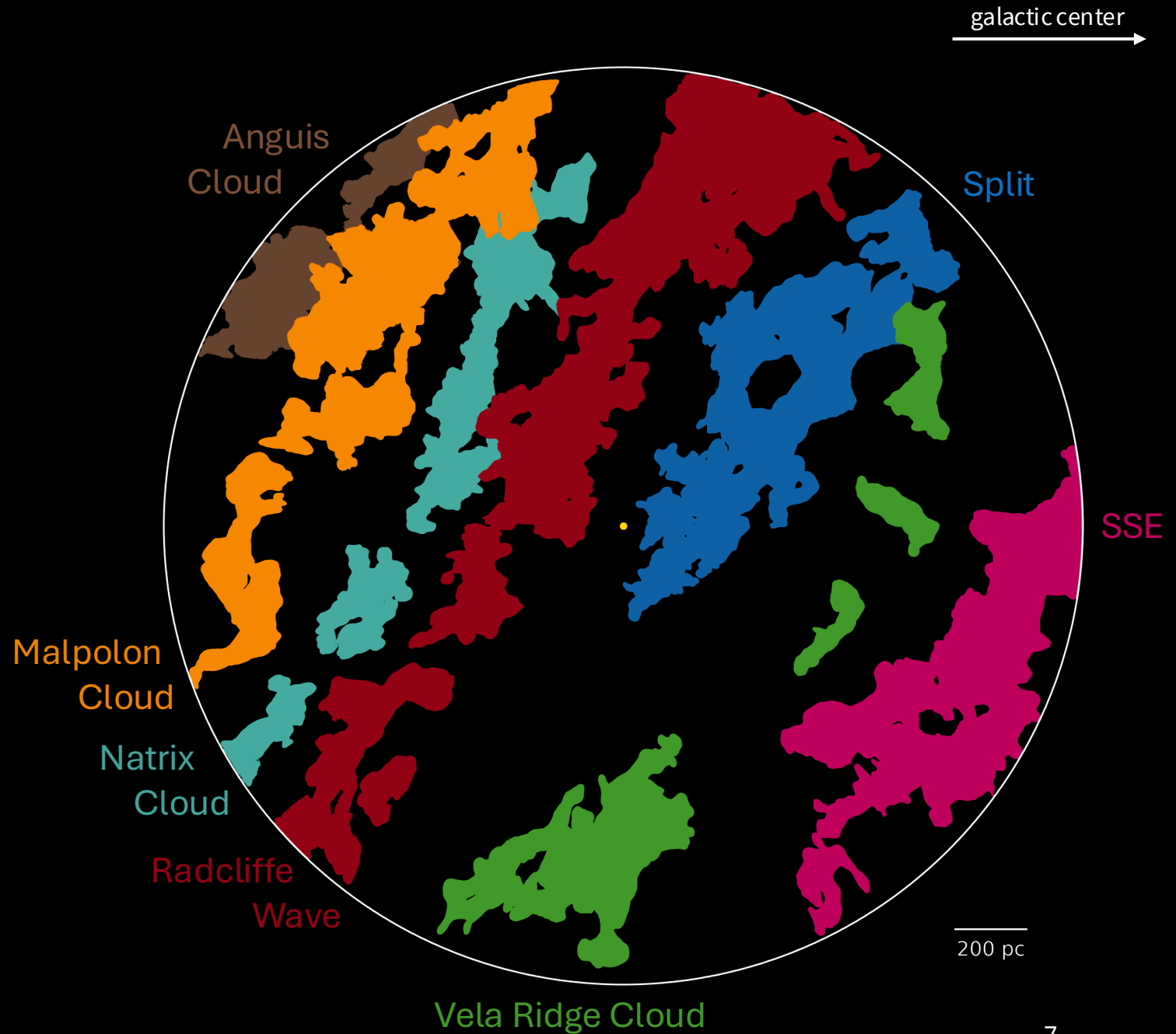
lengths of 1 – 2.5 kpc
masses of $\sim 10^6 M_{\odot}$
 $\sim 30^{\circ}$ pitch angle



Superclouds

7 superclouds of the local
Milky Way

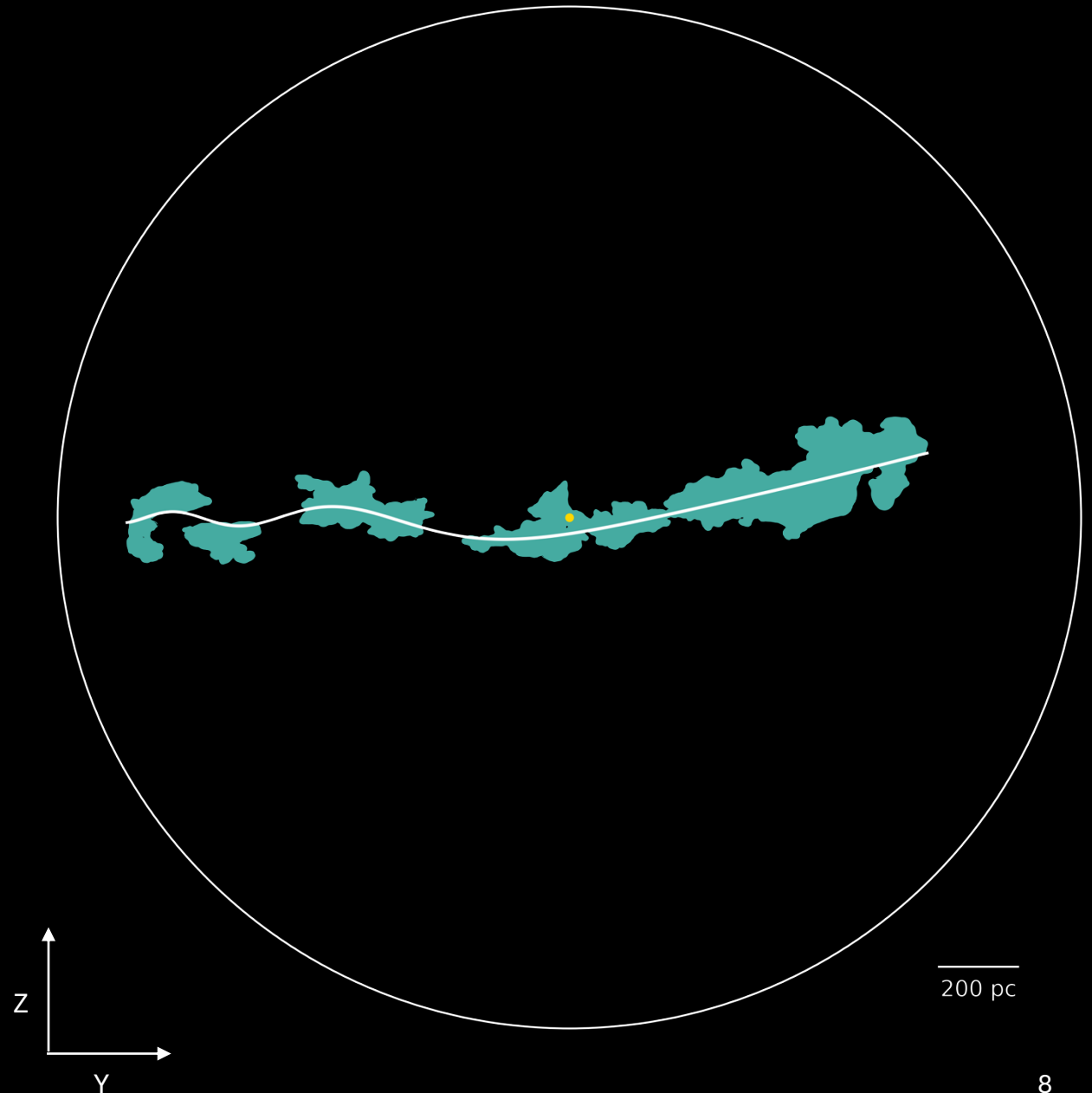
lengths of 1 – 2.5 kpc
masses of $\sim 10^6 M_{\odot}$
 $\sim 30^\circ$ pitch angle



Undulation

superclouds are rotated
by their pitch angle

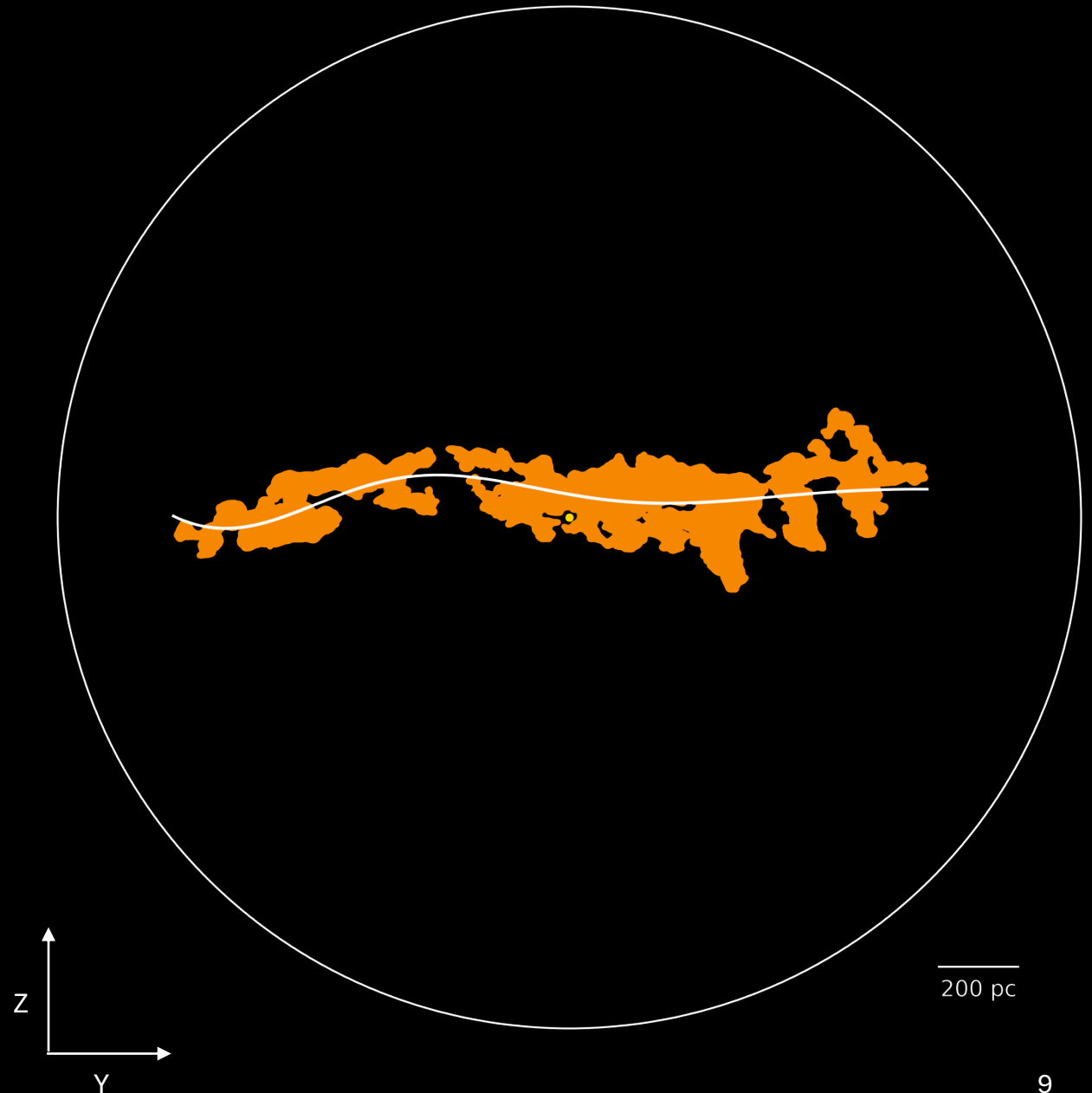
Matrix Cloud



Undulation

superclouds are rotated
by their pitch angle

Malpolon Cloud

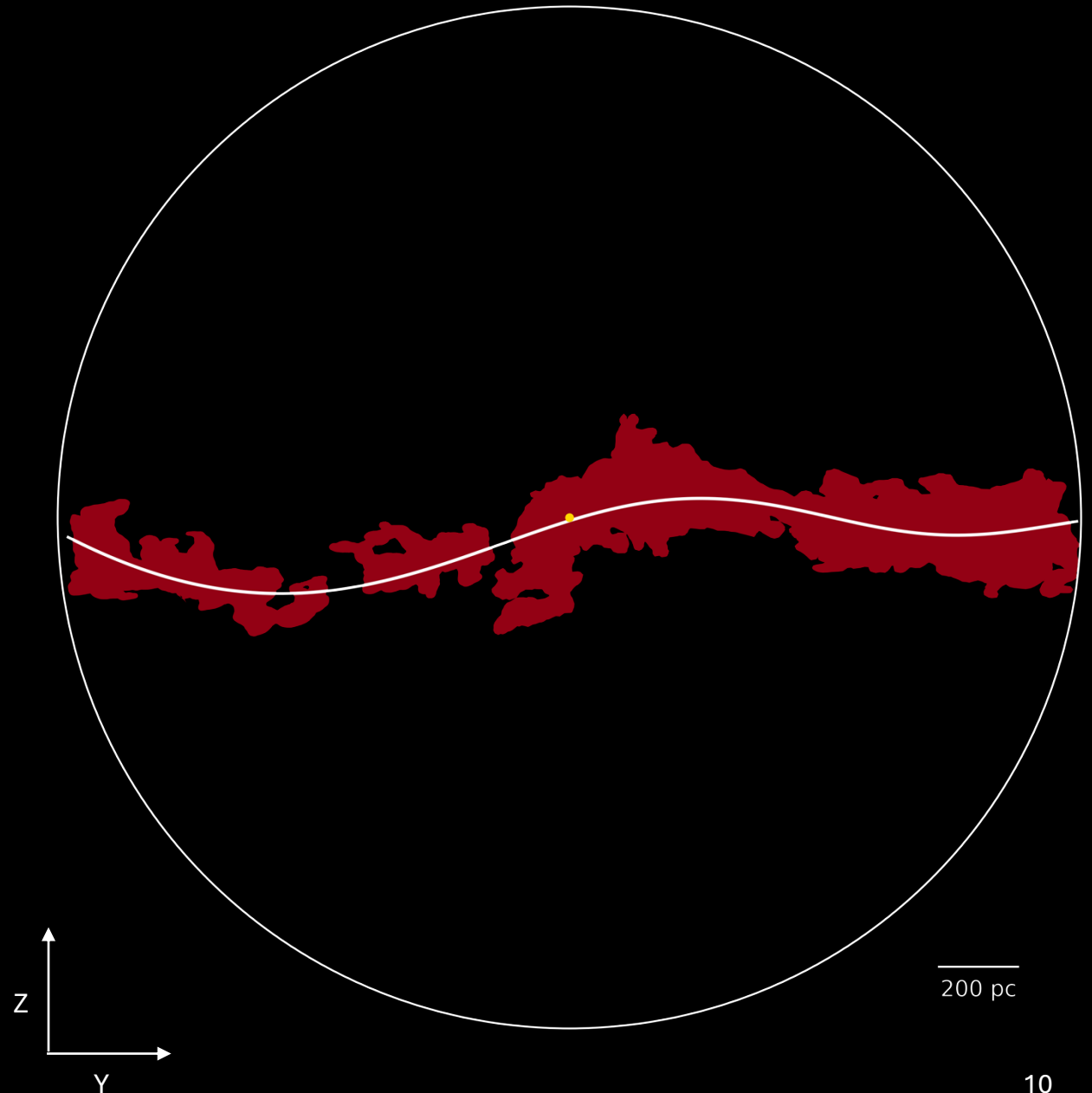


Undulation

superclouds are rotated
by their pitch angle

Radcliffe Wave

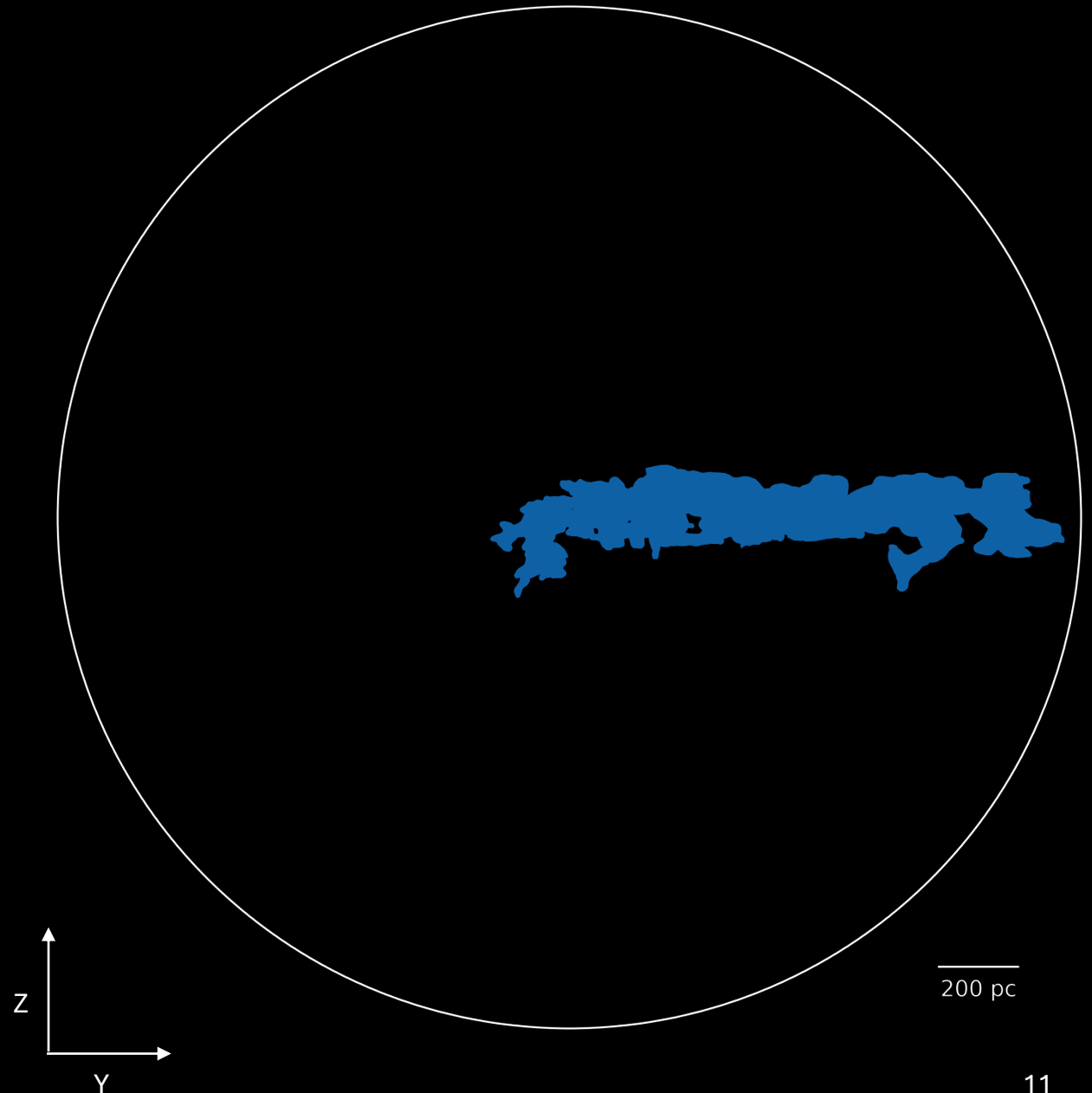
oscillating structure
(Konietzka et al. 2024)



Undulation

superclouds are rotated
by their pitch angle

Split
only supercloud
showing no undulation



Superclouds

90% of SFRs in the dust
map volume inside the
superclouds

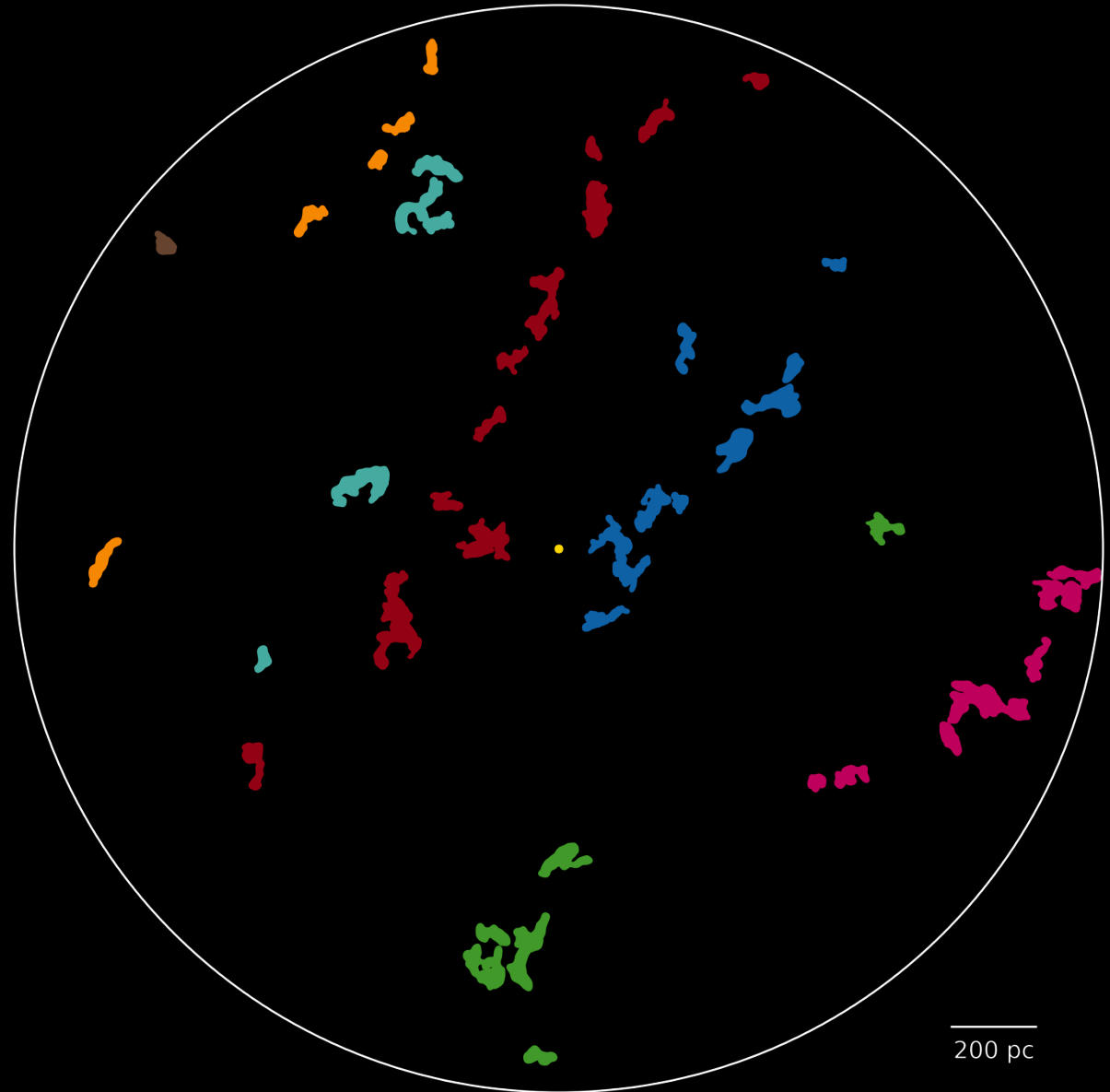
reapplication of HOP
using higher density
parameters



Superclumps

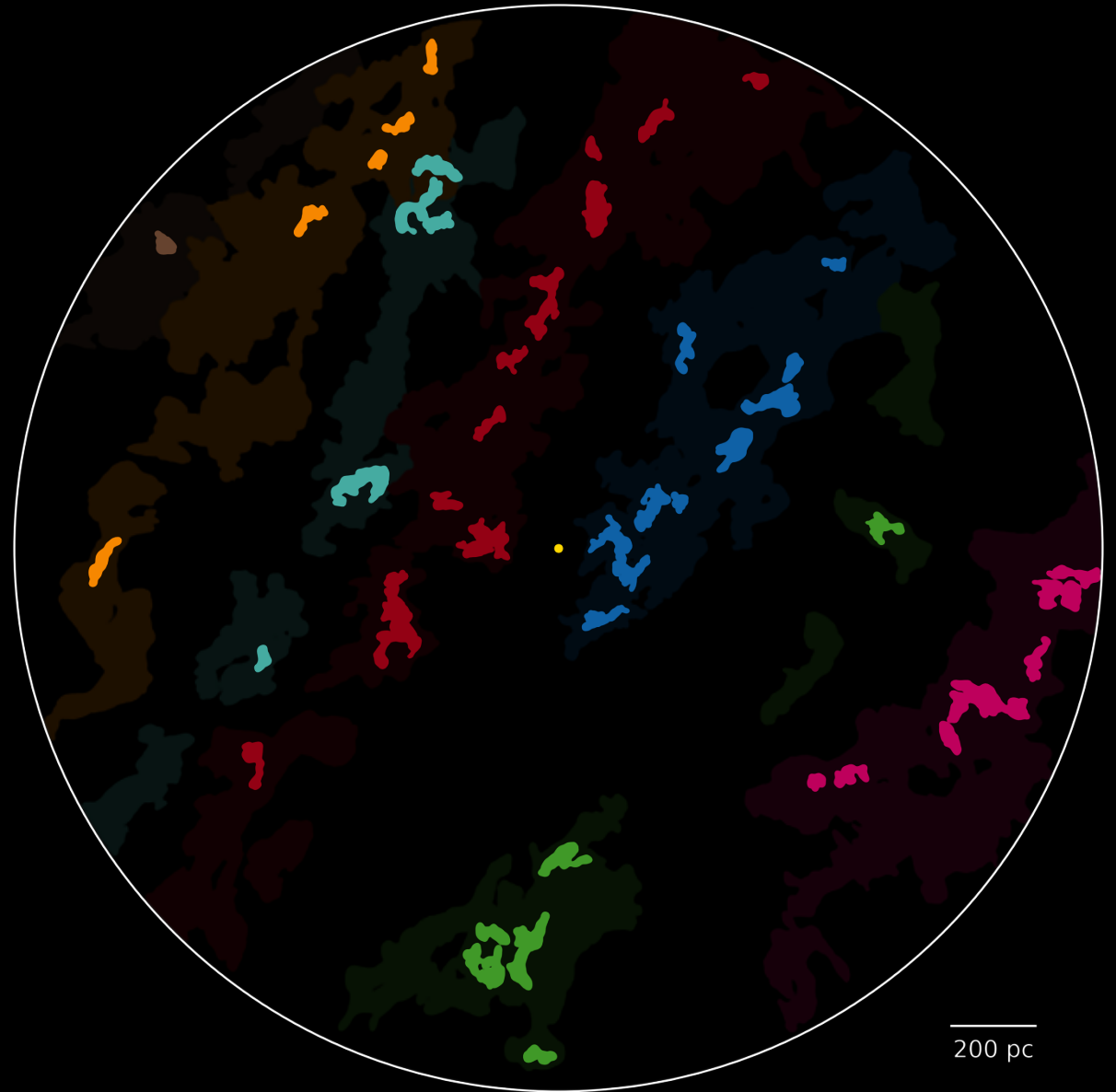
density enhancements
inside the superclouds

lengths of ~ 100 pc
masses of $\sim 10^4 M_{\odot}$
150 – 380 pc spacing



Combined View

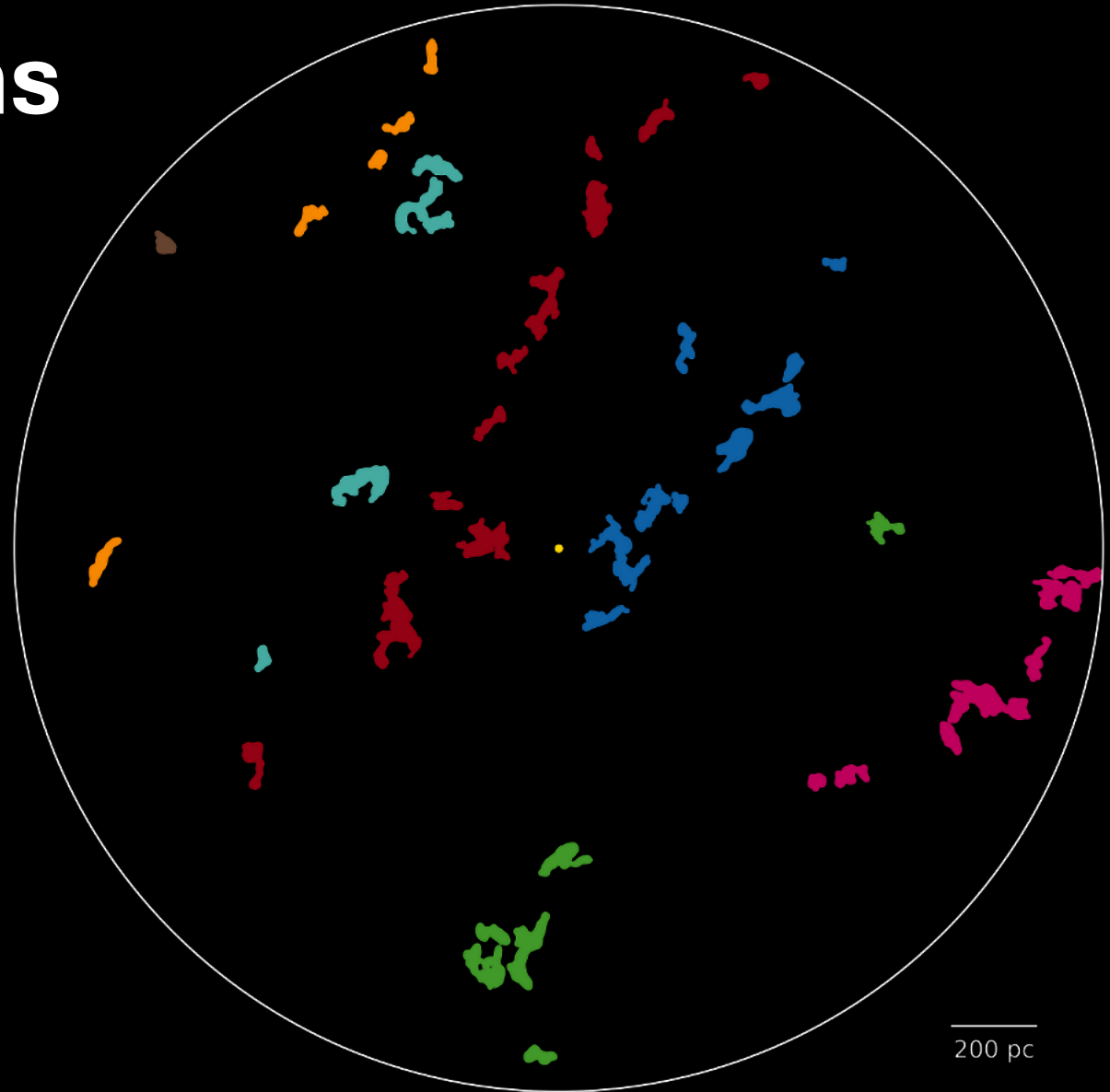
superclumps align with
supercloud spines



Star-Forming Regions

SFRs from Zucker et al.
(2020) within the volume
of the dust map

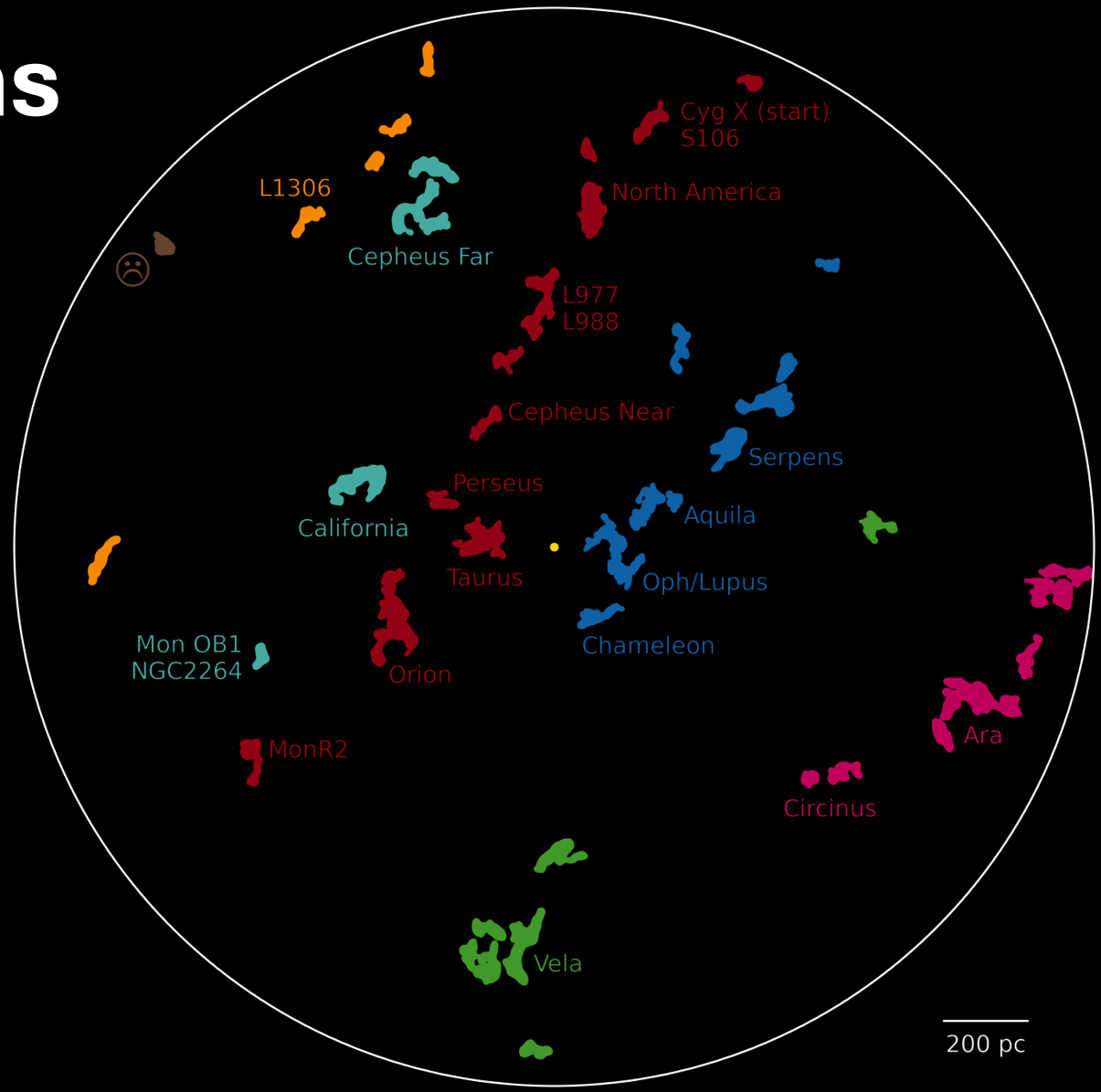
90% within superclouds
73% within superclumps



Star-Forming Regions

SFRs from Zucker et al.
(2020) within the volume
of the dust map

90% within superclouds
73% within superclumps



Conclusion

- identification of 7 superclouds in the local Milky Way with 5 previously unknown structures
- inner density analysis reveals superclumps, $\sim 73\%$ of known SFRs overlap and we find repeated spacings
- superclouds and superclumps act as bridge between galactic scales down to star-forming regions

Superclouds
Kormann+2026



Superclumps
(Preprint)

